Why an O(h) Mean Error Can Produce a Finite-Grid Plateau

Seed-3 phase quantization in the two-chain atomistic convergence study

Continuum Limit Relaxation research audit

September 7, 2026

Audience: Continuum Limit Relaxation research project

Question: Why is the seed-3 mean contribution almost constant from $N = 80$ to $N = 320$ even though Appendix B proves that it is $O(h)$?

Executive answer

There is no contradiction. The estimate $M_N = O(h_N)$ is an upper-envelope statement, not a pointwise halving law. Appendix B leaves a dimensionless nearest-integer remainder q_N that may vary between $-1/2$ and $1/2$:

$$\overline{u}_{1,N} = \frac{h_N}{2} q_N, \qquad M_N := 2\sqrt{\pi}|\overline{u}_{1,N}| = \sqrt{\pi} h_N |q_N|.$$

The bound $|q_N| \leq 1/2$ proves $M_N \leq (\sqrt{\pi}/2)h_N$, but it does not say that q_N is constant or convergent. In the saved seed-3 data, h_N nearly halves while $|q_N|$ nearly doubles over $N = 80, 160, 320$. Those two effects cancel, creating the plateau. At $N = 640$, the nearest-integer remainder wraps across the edge of its normalization cell, q_N drops sharply, and the error falls again.

1. Exact Appendix B formula

Let a_N denote the unnormalized phase quantity

$$a_N := \frac{h_2\overline{u}_2 - h_1\overline{u}_1}{h_1 h_2}.$$

Appendix B chooses $d_N \in \mathbb{Z}$ to be the nearest integer to a_N and then uses the continuous common translation c_N to enforce $\overline{\tilde{u}}_1 + \overline{\tilde{u}}_2 = 0$. The normalized first mean is exactly

$$\overline{\tilde{u}}_{1,N} = \frac{h_1 h_2}{h_1 + h_2}(d_N - a_N) = \frac{h_N}{2} q_N, \qquad q_N := d_N - a_N.$$

Nearest-integer rounding gives $|q_N| \leq 1/2$, hence

$$|\overline{\tilde{u}}_{1,N}| \leq \frac{h_N}{4}.$$

For the two-layer constant error $(m, -m)$, the derivative terms vanish and

$$\|(m, -m)\|_{H^2_h} = 2\sqrt{\pi}|m|.$$

Therefore the plotted mean mode is

$$M_N = \sqrt{\pi} h_N |q_N| \leq \frac{\sqrt{\pi}}{2} h_N.$$

This proves an $O(h)$ envelope but imposes no monotonicity on M_N.

2. What happens in the saved seed-3 sequence

The current output gives the following exact reconstruction. The mean-free column is computed from the orthogonal decomposition $E_N^2 = M_N^2 + (E_N^{\mathrm{mf}})^2$.

N	d_N	h_N	$q_N = 2\overline{u}_1/h_N$	M_N	full H_h^2 error	mean-free error
20	0	0.306497	0.282933	0.153704	0.159089	0.041041
40	-1	0.155140	-0.439450	0.120840	0.124476	0.029867
80	-1	0.078052	0.113199	0.015660	0.016874	0.006283
160	-2	0.039148	0.216269	0.015006	0.015619	0.004331
320	-4	0.019604	0.422538	0.014682	0.015129	0.003647
640	-9	0.009810	-0.165304	0.002874	0.003020	0.000928

From $N = 80$ to 160,

$$\frac{h_{160}}{h_{80}} = 0.5016, \qquad \frac{|q_{160}|}{|q_{80}|} = 1.9105, \qquad \frac{M_{160}}{M_{80}} = 0.9582.$$

From $N = 160$ to 320,

$$\frac{h_{320}}{h_{160}} = 0.5008, \qquad \frac{|q_{320}|}{|q_{160}|} = 1.9538, \qquad \frac{M_{320}}{M_{160}} = 0.9784.$$

Thus the plateau is genuinely almost flat: its local mean-mode slopes are about -0.062 and -0.032. It is not a plotting illusion.

3. Why the remainder nearly doubles

The reconstructed unrounded phase satisfies

$$\frac{a_N}{N} = -0.014147, -0.014014, -0.013915, -0.013852, -0.013820, -0.013804$$

for $N = 20, 40, 80, 160, 320, 640$. Hence a_N is approximately proportional to N: the same physical phase displacement is being expressed in increasingly fine lattice-index units.

On a dyadic refinement this gives $a_{2N} \approx 2a_N$. Since $q_N = \mathrm{round}(a_N) - a_N$, the remainder follows the sawtooth or doubling map

$$q_{2N} \approx 2q_N - \mathrm{round}(2q_N).$$

For $N = 80, 160, 320, 640$, the observed progression is

$$0.1132 \longrightarrow 0.2163 \longrightarrow 0.4225 \longrightarrow -0.1653.$$

Before the wrap, $|q_N|$ nearly doubles. Because h_N nearly halves,

$$M_{2N} \approx \sqrt{\pi}\frac{h_N}{2}(2|q_N|) \approx M_N.$$

Once $2q_N$ crosses the nearest-integer cell boundary, the rounding subtracts an integer and resets the remainder. This produces the sharp drop at $N = 640$. The observed next-grid remainders differ from this simple doubling prediction by only about 0.005 to 0.010.

4. Why this remains O(h)

Big-O means that there are constants C and N_0 such that

$$M_N \le C h_N \qquad (N \ge N_0).$$

It does not require $M_{2N} = M_N/2$, a local log-log slope of -1, or monotone decay. Here

$$\frac{M_N}{h_N} = \sqrt{\pi}|q_N| \le \frac{\sqrt{\pi}}{2}$$

uniformly. A nonzero absolute plateau cannot persist forever, because doing so would force M_N/h_N to exceed this bound as $h_N \to 0$. It can persist for a few refinements while $|q_N|$ grows, then it must drop. Similar, smaller sawteeth may recur at later grids.

Arithmetic sensitivity of convergence rates is also natural in broader incommensurate bilayer limits: Hott, Watson, and Luskin obtain a quantitative rate under a Diophantine condition on the twist [S2]. This does not prove the present rounding formula; the exact explanation here comes from Appendix B and the saved normalization data.

5. Interpretation and next study

The mean explains most, but not all, of the gap between seed 3 and the warm track. After removing the exact mean component, the seed-3 mean-free error is still larger than the warm error at every saved grid. Its six-grid fitted slope is about -1.056, so it is broadly first order but not perfectly straight. This mean-free solution error must not be called the consistency residual: $r_h = \Delta_h u_c - F_h(u_c)$ is a different quantity and is not currently saved.

A focused follow-up should retain the full Theorem 5 error and add:

1. $q_N = 2\overline{u}_1/h_N$ and the selected integer d_N;
2. the exact mean mode M_N;
3. the mean-free H_h^2 error $E_N^{\mathrm{mf}} = \sqrt{E_N^2 - M_N^2}$;
4. the actual consistency residual $\|\Delta_h u_c - F_h(u_c)\|$.

This separates phase quantization, profile-shape error, and consistency. It also avoids an impermissible continuous shift of the atomistic endpoint.

Sources and evidence ledger

[S1] *Continuum Limit of a One-Dimensional Atomistic Model for Moire Relaxation*, July 21, 2026, local manuscript. Used for Theorem 3, Theorem 4, Theorem 5, and Appendix B. Accessed from `Hao_OralExam_Continuum_Limit.pdf` in the project repository.

[D1] Current atomistic convergence-study implementation and saved `convergence_summary.csv`, September 7, 2026. Used for d_N, normalized means, errors, and the verified sawtooth reconstruction.

[S2] Michael Hott, Alexander B. Watson, and Mitchell Luskin, *From Incommensurate Bilayer Heterostructures to Allen-Cahn: An Exact Thermodynamic Limit*, arXiv:2305.18186, May 29, 2023. Used only as broader context for arithmetic conditions in quantitative incommensurate convergence rates.

Limitations. The six saved resolutions verify the mechanism for the current seed-3 track, but do not prove that the same sawtooth timing persists for all N. The optimizer endpoints are accepted stationary candidates, not certified global minimizers, and the manuscript's uniform stability gap is not checked numerically.